Group II Row IV: from the three lines to JIMWLK, BFKL, DGLAP, running coupling and finite terms

1 Notation

- $d_\perp = 2-\epsilon$; $a \equiv \frac{k^+}{\vee}$; $X \equiv y'-y$; $c_0 = 2e^{-\gamma_E}$; $\hat{\mu}^2 \equiv \frac{\mu_{\overline{MS}}^2 e^{2\gamma_E}}{4} = \frac{\mu_{\overline{MS}}^2}{c_0^2}$; $\beta_0 = \frac{11}{3}N_c$.
- $c \equiv \frac{g^2 N_c}{4\pi^2} = \frac{g^2}{8\pi^2}\,2N_c = \frac{\alpha_s}{2\pi}2N_c$; $\mathcal{K}(r) \equiv \frac{r}{r^2}$.
- LO with a weight $f(X)$ *inside* the integral, and its rescaled version:

$$\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\, f(y'-y)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} \times \big[U^{e'c}(y') - U^{e'c}(x')\big]\big[U^{ec}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \qquad \frac{d^3N_{\mathrm{LO}}}{d^3k}[f] \equiv \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[f]\Big|_{\xi=1}. \tag{1.1}$$

- $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\mathrm{lp}}$: Eq. (1.1) with $f=1$ and the kernel of line 3.
- The point-z LO (both gluon points at z, no phase) and H_4:

$$\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y} \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,x',z} \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{1.2}$$

$$H_4 \equiv \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \times \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \tag{1.3}$$

2 The starting point

Your Row IV, the three lines (L1), (L2), (L3):

$$\begin{aligned}
(\mathrm{L1}) &= c\int_a^1 \frac{d\xi}{\xi^2}\Big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\Big]\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}_0],\\
(\mathrm{L2}) &= c\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big)\Big]\frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}_0],\\
(\mathrm{L3}) &= c\int_a^{\frac{k^+}{k^++\Lambda}} d\xi\,\frac{1-\xi}{\xi}\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\mathrm{lp}}\Big|_{d_\perp},
\end{aligned} \tag{2.1}$$

with

$$\mathcal{B}_0 \equiv -\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{X^2\mu_{\overline{MS}}^2}{4} - 1. \tag{2.2}$$

The bracket depends on X, so it is a weight inside the integrals.

3 Step 0: what line 2 is

Put $p^+ = k^+\frac{1-\xi}{\xi}$. Then $\xi = 1 \leftrightarrow p^+ = 0$, $\xi = a \leftrightarrow p^+ = \vee - k^+$, $\xi = \frac{k^+}{k^++\Lambda} \leftrightarrow p^+ = \Lambda$, and

$$\frac{d\xi}{\xi^2}\,\frac{\xi}{1-\xi} = \frac{dp^+}{p^+}. \tag{3.1}$$

Write $F(\xi) \equiv \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}_0]$, so $F(1) = \frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}_0]$. The $\frac{\xi}{(1-\xi)_+}$ term of (L1) is your p^+ prescription:

$$\int_a^1 \frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}F \equiv \int_a^1 d\xi\,\frac{F(\xi)-F(1)}{\xi(1-\xi)} = \int_0^{\vee-k^+} dp^+\,\frac{F(p^+)-F(0)}{p^+}. \tag{3.2}$$

The unregulated integral with the cut $p^+ \geq \Lambda$ is, exactly,

$$\int_\Lambda^{\vee - k^+} \frac{dp^+}{p^+} F = \int_0^{\vee - k^+} dp^+ \frac{F(p^+) - F(0)}{p^+} + F(0)\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+} - 1\Big)\Big] + O\Big(\frac{\Lambda}{k^+}\Big). \tag{3.3}$$

So (L2) is the subtraction term of your plus prescription: (L1)+(L2) is the original integral over $p^+ \geq \Lambda$. (L2) has the LO Wilson-line structure because $F(0) = \frac{d^3 N_{\rm LO}}{d^3 k}[\mathcal{B}_0]$. (Checked: 4×10^{-16}, `g2iv_full/plus_check.py`.)

4 Step 1: the ϵ-bracket

With $d_\perp = 2 - \epsilon$, $-\frac{d_\perp}{\epsilon} = -\frac{2}{\epsilon} + 1$, and the -1 cancels. Exactly:

$$\mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\Big[2\gamma_E + \log\frac{X^2\mu_{\overline{MS}}^2}{4}\Big] = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ln\big(X^2\hat{\mu}^2\big) \equiv \mathcal{B}. \tag{4.1}$$

Where no pole multiplies it, $\frac{d_\perp}{2} = 1 - \frac{\epsilon}{2} \to 1$.

5 Step 2: standard plus distribution; the $\log(\frac{\vee}{k^+} - 1)$ cancels

DGLAP uses the standard plus distribution on $[0,1]$, restricted to $[a,1]$:

$$\int_a^1 \frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}\Big|_{\rm std} F \equiv \int_a^1 d\xi\, \frac{F(\xi)/\xi - F(1)}{1-\xi} + F(1)\log(1-a). \tag{5.1}$$

Every term here is finite. $\int_a^1 \frac{d\xi}{1-\xi}$ alone diverges, but it never appears alone:

- In the integral the numerator $F(\xi)/\xi - F(1) \to 0$ linearly as $\xi \to 1$, so the integrand is bounded at $\xi = 1$.
- $\log(1-a) = \log(1 - \frac{k^+}{\vee})$ is finite because $a < 1$. It does not come from $\int_a^1$; it comes from $\int_0^a \frac{d\xi}{1-\xi} = -\log(1-a)$.

Derivation. Write $f(\xi) \equiv F(\xi)/\xi$, so $\frac{d\xi}{\xi^2}\,\xi\,F = d\xi\, f$.

1. *Definition.* $\frac{1}{(1-\xi)_+}$ is defined on $[0,1]$ by $\int_0^1 d\xi \frac{g(\xi)}{(1-\xi)_+} = \int_0^1 d\xi \frac{g(\xi)-g(1)}{1-\xi}$. The lower limit a is a $\theta(\xi - a)$ inside g: $g(\xi) = \theta(\xi-a)f(\xi)$, with $g(1) = f(1)$. Then

$$\int_0^1 d\xi \frac{\theta(\xi-a)f(\xi) - f(1)}{1-\xi} = \int_a^1 d\xi \frac{f(\xi)-f(1)}{1-\xi} - f(1)\int_0^a \frac{d\xi}{1-\xi} = \int_a^1 d\xi \frac{f(\xi)-f(1)}{1-\xi} + f(1)\log(1-a),$$

   which is Eq. (5.1), since $f(1) = F(1)$.

2. *Same thing with an explicit cut.* Equivalently, $\frac{1}{(1-\xi)_+} = \lim_{\delta\to 0}\big[\frac{\theta(1-\delta-\xi)}{1-\xi} + \log\delta\,\delta(1-\xi)\big]$. Then

$$\int_a^{1-\delta} d\xi \frac{f}{1-\xi} + f(1)\log\delta = \int_a^{1-\delta} d\xi \frac{f-f(1)}{1-\xi} + f(1)\Big[\log\frac{1-a}{\delta} + \log\delta\Big] \xrightarrow{\delta\to 0} \int_a^1 d\xi \frac{f-f(1)}{1-\xi} + f(1)\log(1-a).$$

   The divergent $\log\delta$ of the cut integral cancels against the $\delta(1-\xi)$ term (as with the $\delta(1-\xi)$ of $\mathcal{P}_{gg}$, virtual against real).

Numerical check (`advisor/plus_finite.py`, $a = 10^{-2}$): the cut integral grows as 1.71, 3.25, 4.78, 6.32 for $\delta = 10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$. Adding $f(1)\log\delta$ gives 0.18020, 0.184779, 0.1848243, 0.18482476, approaching Eq. (5.1) = 0.18482477.

The difference from yours, Eq. (3.2), combining the integrands before integrating, so that nothing divergent is split off:

$$\begin{aligned}\text{yours} - \text{std} &= \int_a^1 d\xi\Big[\frac{F-F(1)}{\xi(1-\xi)} - \frac{F/\xi - F(1)}{1-\xi}\Big] - F(1)\log(1-a) = F(1)\int_a^1 d\xi\,\frac{-1+\xi}{\xi(1-\xi)} - F(1)\log(1-a)\\ &= -F(1)\int_a^1 \frac{d\xi}{\xi} - F(1)\log(1-a) = -F(1)\Big[\log\frac{\vee}{k^+} + \log\Big(1 - \frac{k^+}{\vee}\Big)\Big] = -F(1)\log\Big(\frac{\vee}{k^+} - 1\Big)\end{aligned} \tag{5.2}$$

(checked: 6×10^{-17}). Define the standard splitting function

$$\mathcal{P}_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+}\Big|_{\rm std} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big] + \frac{\beta_0}{2}\delta(1-\xi), \tag{5.3}$$

with $\delta(1-\xi)$ at the endpoint counted fully: $\int_a^1 \frac{d\xi}{\xi^2}\delta(1-\xi)F = F(1)$. Since $c = \frac{g^2}{8\pi^2}2N_c$:

$$\text{(L1)} = \frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg}(\xi) - \frac{\beta_0}{2}\delta(1-\xi)\Big]\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}] \;-\; c\log\Big(\frac{\vee}{k^+}-1\Big)\frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}]. \tag{5.4}$$

Adding (L2), the $\log(\frac{\vee}{k^+}-1)$ cancels exactly (both its $\log\frac{\vee}{k^+}$ and its $\log(1-\frac{k^+}{\vee})$):

$$\boxed{\text{(L1)} + \text{(L2)} = \underbrace{c\,\log\frac{k^+}{\Lambda}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}]}_{\text{JIMWLK}} + \frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg}(\xi) - \frac{\beta_0}{2}\delta(1-\xi)\Big]\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}]} \tag{5.5}$$

- The $\log\frac{\vee}{k^+}$ with the LO Wilson-line structure is *not* a BFKL term: it is gone. (In your split it was the pair $\pm c\log\frac{\vee}{k^+}\frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}]$.)
- The $\log\frac{k^+}{\Lambda}$ term is the JIMWLK term of this row, as you say.

6 Step 3: DGLAP and running coupling

Split Eq. (5.5) at $\frac{\beta_0}{2}\delta(1-\xi)$:

$$\underbrace{-\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon} + \frac{d_\perp}{2}\ln(X^2\hat{\mu}^2)\Big]}_{\text{(D)}} \;+\; \underbrace{\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon} + \frac{d_\perp}{2}\ln(X^2\hat{\mu}^2)\Big]}_{\text{(R)}}. \tag{6.1}$$

DGLAP. The hadron is $\frac{d^3N_h}{d^3k} = \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3}D_{h/g}(\zeta)\frac{d^3N_g}{d^3k}\big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\big)$, and $D^{\mathrm{bare}} = D(\mu_F) + \frac{g^2}{8\pi^2}\big[\frac{2}{\epsilon} + \ln\frac{\mu^2}{\mu_F^2}\big]\mathcal{P}_{gg}\otimes D$.

- The counterterm covers the same region as the ξ integral ($a \le \zeta\xi \le \zeta \le 1$), with the same standard plus distribution. So the pole of (D) cancels.
- What is left is

$$\text{(D)} \to -\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\Big[\ln\frac{X^2\mu_F^2}{c_0^2}\Big]. \tag{6.2}$$

Running coupling. $\frac{d^3N_{\mathrm{LO}}}{d^3k} \propto g^2$, and the $\overline{MS}$ coupling counterterm adds $-\frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\frac{d^3N_{\mathrm{LO}}}{d^3k}$:

$$\text{(R)} - \frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\frac{d^3N_{\mathrm{LO}}}{d^3k} = \frac{\alpha_s\beta_0}{4\pi}\frac{d^3N_{\mathrm{LO}}}{d^3k}\big[\ln(X^2\hat{\mu}^2)\big], \qquad \frac{d^3N_{\mathrm{LO}}}{d^3k}\Big|_{\alpha_s} + \frac{\alpha_s\beta_0}{4\pi}\frac{d^3N_{\mathrm{LO}}}{d^3k}\big[\ln(X^2\hat{\mu}^2)\big] = \frac{d^3N_{\mathrm{LO}}}{d^3k}\Big|_{\alpha_s\to\alpha_s(X^2)}, \tag{6.3}$$

with $\alpha_s(X^2) = \alpha_s + \frac{\alpha_s^2\beta_0}{4\pi}\ln(X^2\hat{\mu}^2) + O(\alpha_s^3)$ and $X = y' - y$ (checked, sympy).

7 Step 4: the BFKL term hidden in (D)

At small ξ, $\mathcal{P}_{gg} \to 2N_c\frac{1-\xi}{\xi}$, and the y' integral is a Fourier transform at $\mathbf{k}/\xi$. Write the integrand of $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[f]$ as $e^{-ik\cdot(y'-y)/\xi}f(y'-y)\,F_y(y')$, with F_y smooth.

1. The constants and $\ln\mu^2$ in the weight multiply $\int d^2y' e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^2(\mathbf{k}) = 0$ for $\mathbf{k}\neq 0$.
2. $\nabla^2\ln X^2 = 4\pi\delta^2(X)$ gives $\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\ln(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2}$ (checked).
3. Hence $\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\ln(X^2\hat{\mu}^2)F_y(y') = -\frac{4\pi\xi^2}{\mathbf{k}^2}F_y(y)\big[1+O(\xi)\big]$. Numerically (`advisor/smallxi.py`) the ratio to the leading term is 1.008, 1.002, 0.99994, 0.99979 at $\xi = 0.2, 0.1, 0.05, 0.03$.
4. At $y' = y \equiv z$: $\frac{(x'-y')}{(x'-y')^2} \to \mathcal{K}(x'-z)$ and $U(y') \to U(z)$. So $F_y(y)$ is the integrand of $\frac{d^3N_{\mathrm{LO}}}{d^3k}^y$, with the Wilson-line structure $[U(z)-U(x')][U(z)-U(x)]$.

The small-ξ end of (D) is therefore

$$-\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\frac{d^3N_{\mathrm{LO}}}{d^3k}^y = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N_{\mathrm{LO}}}{d^3k}^y\int_a^1 d\xi\,\frac{1-\xi}{\xi} = \Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]H_4, \tag{7.1}$$

since $\int_a^1\frac{1-\xi}{\xi}d\xi = \log\frac{1}{a} - (1-a)$ and $H_4 = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N_{\mathrm{LO}}}{d^3k}^y$. The last identity holds because $\int_w e^{ik\cdot w}\mathcal{K}^i(w-z) = \frac{2\pi ik^i}{\mathbf{k}^2}e^{ik\cdot z}$, so $\int_{w,w'} e^{-ik\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z) = \frac{4\pi^2}{\mathbf{k}^2}$ (finite, no $\log\mu$; checked).

- Expanding $[U(z)-U(x')][U(z)-U(x)]\rho\rho$ gives exactly your four-line BFKL term ($\frac{g^4}{16\pi^5}\cdot 2N_c = \frac{g^4N_c}{8\pi^5}$).
- **This is where your BFKL term comes from**, with the right Wilson-line structure. It is the parent gluon with large p^+ (BFKL region) inside the DGLAP integral, and no other row has this structure.

8 Step 5: extract it

Add and subtract Eq. (7.1) inside (D):

$$(\mathrm{D}) = \underbrace{-\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big]+\frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}^{\,y}\Big\}}_{(\mathrm{D_{sub}})\text{: DGLAP with the BFKL region subtracted}}$$
$$\underbrace{+\log\frac{\vee}{k^+}H_4}_{\to\ \mathrm{BFKL}}\ \underbrace{-\Big(1-\frac{k^+}{\vee}\Big)H_4}_{\text{finite}}. \tag{8.1}$$

- By Step 4 the integrand of $(\mathrm{D_{sub}})$ is integrable at $\xi\to 0$, so $(\mathrm{D_{sub}})$ has no $\log\frac{\vee}{k^+}$.
- The subtraction has no pole and no μ: the DGLAP renormalization of Step 3 is unchanged.
- Everything is symmetric between amplitude and conjugate. $\frac{d^3N_{\mathrm{LO}}}{d^3k}^{\,y}$ has both gluon points at one z.
- This replaces the $-2N_c\frac{1-\xi}{\xi}\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}[\log\dots]$ of the earlier Eq. (5.12), which as written (log outside) is zero.

9 Step 6: line 3

$\int_a^{\frac{k^+}{k^++\Lambda}} = \int_a^1 + O(\frac{\Lambda}{k^+})$, since the integrand $\propto(1-\xi)$. At small ξ, $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\,\mathrm{lp}} = O(\xi^2)$ (no $\delta^2(\mathbf{k})$ term), so there is no $\log\frac{\vee}{k^+}$. No pole multiplies it, so $d_\perp\to 2$:

$$(\mathrm{L3}) = c\int_a^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}\dots\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-2\frac{(x-y)\cdot X\,(x'-y')\cdot X}{(x-y)^2(x'-y')^2X^2}\Big]\dots\quad\text{(finite)}. \tag{9.1}$$

10 Result

$$\begin{aligned}\text{Row IV} = {}& \underbrace{c\log\frac{k^+}{\Lambda}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}\Big[-\frac{2}{\epsilon}-\ln(X^2\hat\mu^2)\Big]}_{\text{JIMWLK}}+\underbrace{\log\frac{\vee}{k^+}H_4}_{\text{BFKL}}\\ &-\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big]+\frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}^{\,y}\Big\}\quad\text{(DGLAP)}\\ &+\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big]\quad\text{(running coupling)}\\ &-\Big(1-\frac{k^+}{\vee}\Big)H_4+c\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\,\mathrm{lp}}\Big|_{d_\perp=2}\quad\text{(finite)}\end{aligned} \tag{10.1}$$

After renormalization: DGLAP $\to\ln\frac{X^2\mu_F^2}{c_0^2}$ (pole in $D_{h/g}$), and running coupling $\to\frac{d^3N_{\mathrm{LO}}}{d^3k}|_{\alpha_s\to\alpha_s(X^2)}$.

Check. The three H_4 structures (BFKL, subtraction, finite) add up to zero:

$$\log\tfrac{\vee}{k^+}H_4-\big[\log\tfrac{\vee}{k^+}-(1-a)\big]H_4-(1-a)H_4=0,$$

so the result returns to the starting lines (L1)+(L2)+(L3).

piece of your three lines	goes to	Wilson-line structure
$\log\frac{k^+}{\Lambda}$ of (L2)	JIMWLK	LO
$\log(\frac{\vee}{k^+}-1)$ of (L2)	cancels with your plus prescription (Step 2)	LO
(L1), small-ξ end	BFKL: $+\log\frac{\vee}{k^+}H_4$	$[U(z)-U(x')][U(z)-U(x)]$ (your term)
(L1), rest	DGLAP + running coupling; $-(1-\frac{k^+}{\vee})H_4$ finite	LO_ξ
(L3)	finite (linear polarization)	LO_ξ

At leading power ($a\to 0$): $\int_a^1\to\int_0^1$, $(1-a)H_4\to H_4$. Checks: `g2iv_full/plus_check.py`, `advisor/checks.py`, `advisor/smallxi.py`.